

C360-Mesh: Temporal, Evidence-Grounded Agentic Customer 360

One-Page Findings and Proposed Approach

Problem framing. The goal is not another passive Customer-360 dashboard or a prompt-driven chatbot. The system must continuously ingest asynchronous customer signals, maintain evolving per-customer context, and commit to a bounded intervention or explicit NO_ACTION, while preserving explainability, traceability, deterministic guardrails and real human approval for risky actions.

Proposed system

C360-Mesh uses a state-centric hybrid multi-agent design. Incoming usage, support, transaction and KYC events are normalized through a schema adapter with event-time, ingestion-time, deduplication and late-event handling. Specialized processors then convert raw signals into typed **Evidence Contracts** containing a claim, confidence, evidence IDs, timestamps and validity. Agents publish these structured conclusions to a shared Customer State Board rather than exchanging unrestricted natural-language reasoning.

Memory and temporal reasoning. The board is backed by working memory, per-customer episodic memory and shared semantic/policy memory. A Temporal Truth Maintenance layer keeps valid-from/valid-until intervals and supersession links, so a late event can revise the current state without erasing what the system believed earlier. A cross-signal Life-Event Agent maintains persistent hypotheses such as relocation or home purchase with evidence and confidence.

Adaptive coordination and decisioning. A Synthesis Agent builds the current customer narrative from structured evidence and relevant memory. Debate is not the default: a verifier/debate path is activated only when agents materially disagree or confidence is low. The system generates a bounded action set (including NO_ACTION), then scores candidates using expected benefit, action cost, eligibility/policy risk and uncertainty. This separates semantic reasoning from deterministic business constraints and creates a falsifiable final decision.

Safety and explainability. Before high-impact actions, an evidence-ablation stability check removes the strongest supporting signal and recomputes the action; fragile decisions are escalated. Risk/ambiguity then routes the flow to controlled autonomy, HITL approval/modify/reject, or a hard-stop specialist queue. Every event, retrieval, tool call, finding, decision and human override is recorded in one audit trace.

Why this architecture

Multi-agent control	Research on MAS failures motivates typed handoffs, scoped state and explicit verification instead of free-form agent chat.
Streaming correctness	Event time and lateness concepts make the architecture robust to asynchronous and out-of-order streams.
Memory freshness	Temporal memory retains evolving facts and historical relationships rather than relying on static vector retrieval alone.
Action quality	Utility/uplift thinking asks whether an intervention is useful and cost-effective, not merely whether churn risk is high.

Implementation direction: Python + FastAPI; lightweight stream/event-log layer; PostgreSQL + pgvector for structured and semantic memory; LangGraph-style persisted workflow/HITL; deterministic/ML specialist processors; minimal dashboard; replay-based evaluation harness.

Research anchors: Cemri et al., “Why Do Multi-Agent LLM Systems Fail?” (2025); Rasmussen et al., “Zep: A Temporal Knowledge Graph Architecture for Agent Memory” (2025); Apache Flink event-time/watermark documentation; Moraes et al., “Uplift Modeling: from Causal Inference to Personalization” (2023); LangGraph interrupt/persistence documentation.